

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness lets agents autonomously regulate knowledge utilization through situational special tokens.

84.33%

Llama-8B ALFWorld All
15.01% Know%

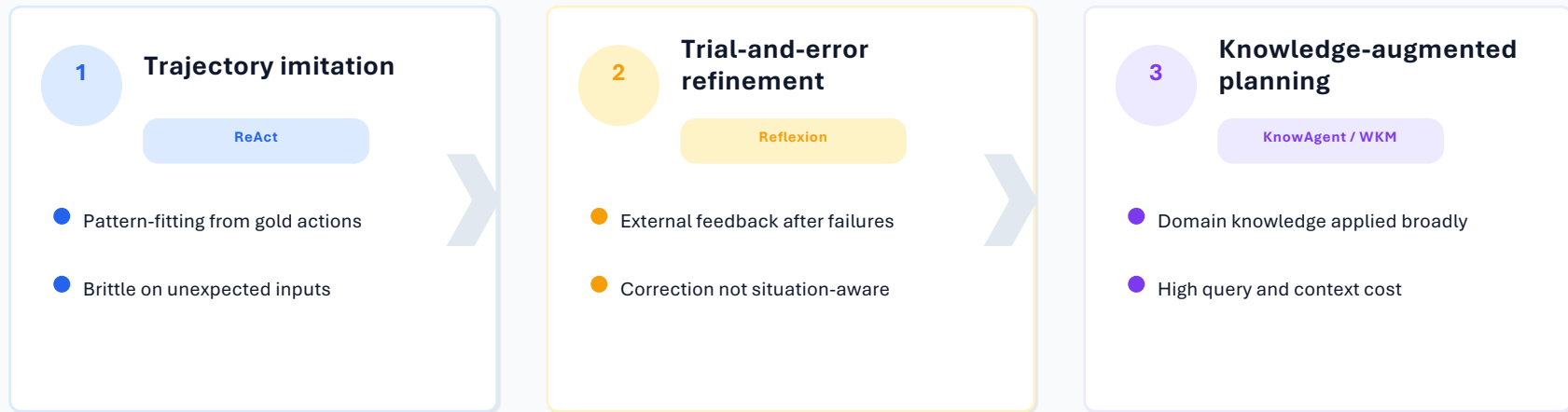
79.85%

Gemma-2B ALFWorld All
26.41% Know%

Source: arxiv.org/abs/2504.03553

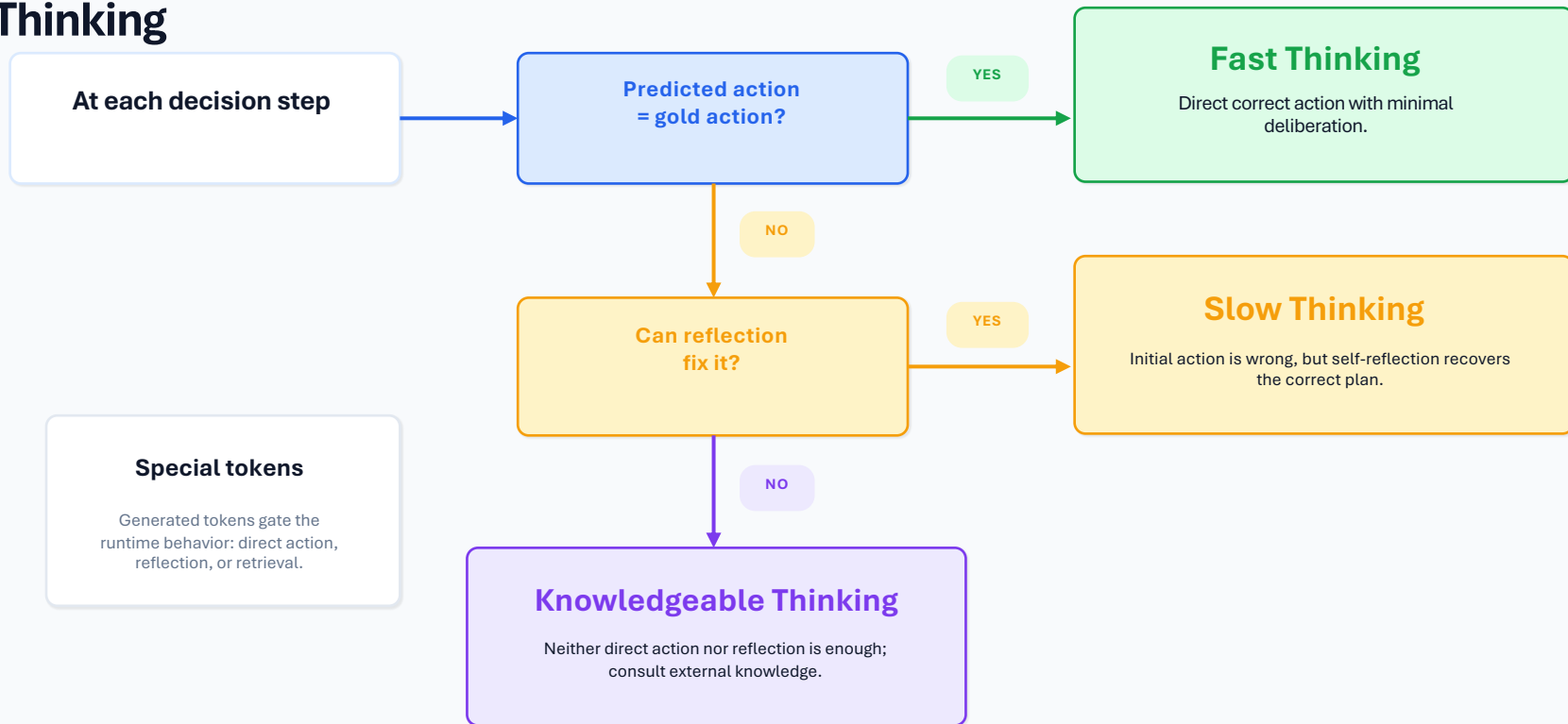
Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

Gold trajectories, external feedback, and domain knowledge are injected regardless of situational need — increasing fragility and inference cost.

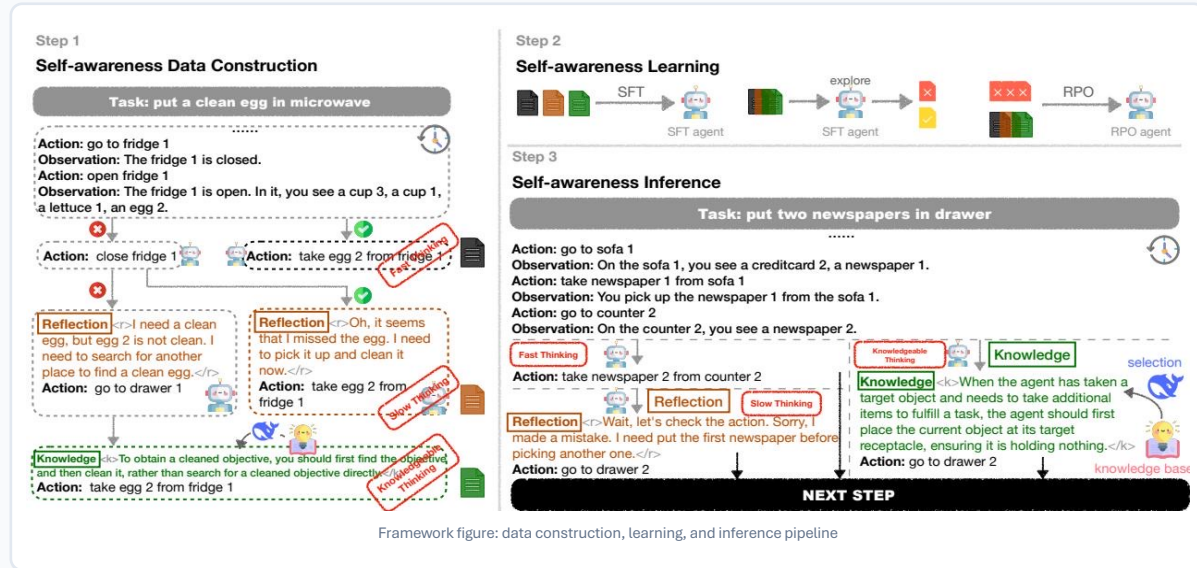


KnowSelf target: replace blanket injection with situational self-awareness — act directly, reflect, or retrieve only when needed.

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference



1 · Self-awareness data construction

- Self-explored trajectories
- Heuristic situation judgment
- Special tokens mark modes

2 · Two-stage training

- SFT teaches initial patterns
- RPO refines token choice
- Optimizes when to trigger tools

3 · Self-aware inference

- Tokens trigger direct action
- Reflection when recoverable
- Knowledge retrieval only when needed

Core mechanism: the policy learns when to emit situation tokens, not merely how to imitate actions.

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

(a) Task: put a hot mug in cabinet

Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
[Knowledge] <|> If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</>
Action: take egg 1 from microwave 1
Observation: Nothing happens.

OpenAI-O1

Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
Action: put mug 1 in/on microwave 1
[Reflection] <|> Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</>
Action: heat mug 1 with microwave 1

KnowSelf
(Llama-8B)

(b) Task: put a saltshaker in drawer

Action: take saltshaker 1 from countertop 3
Observation: You pick up the saltshaker 1 from the countertop 3.
Action: go to drawer 1
Observation: The drawer 1 is open. In it, you see a fork 1.
Action: put saltshaker 1 in/on drawer 1
[Reflection] <|> Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</>
Action: close drawer 1
Observation: You close the drawer 1.
Action: put saltshaker 1 in/on drawer 1
Observation: Nothing happens.

DeepSeek-R1

Action: take saltshaker 1 from countertop 3
Observation: You pick up the saltshaker 1 from the countertop 3.
Action: go to drawer 1
Observation: The drawer 1 is closed.
Action: open drawer 1
Observation: You open the drawer 1. In it, you see a fork 1.
[Knowledge] <|> When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</>
Action: put saltshaker 1 in/on drawer 1

KnowSelf
(Llama-8B)

Qualitative comparison of KnowSelf vs. OpenAI-O1 and DeepSeek-R1

SLOW THINKING · REFLECTION TOKEN

Task: put a hot mug in cabinet

OpenAI-O1 incorrectly tries to take an egg from the microwave. KnowSelf reflects and heats the mug using the open microwave.

KNOWLEDGEABLE THINKING · KNOWLEDGE TOKEN

Task: put a saltshaker in drawer

DeepSeek-R1 gets stuck trying to close an open drawer. KnowSelf retrieves knowledge: place the object without removing unrelated items.

Qualitative pattern: KnowSelf intervenes only when the decision step demands reflection or external knowledge.

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Success rates (%) across six ALFWorld task types. Know% is the proportion of steps using external knowledge.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Readout

84.33%

KnowSelf Llama-8B
15.01% Know%

79.85%

KnowSelf Gemma-2B
26.41% Know%

Key comparison: Llama-8B KnowSelf surpasses GPT-4o ExpeL (79.85%) while using only 15.01% knowledge.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods

Accuracy bars (dark green) are plotted against knowledge usage bars (orange). KnowSelf reaches top scores without blanket retrieval.

Accuracy vs. knowledge usage (%)



Efficiency advantage

15.01% Llama-8B KnowSelf knowledge usage

26.41% Gemma-2B KnowSelf knowledge usage

100% KnowAgent / WKM / ExpeL knowledge usage

79.85% Gemma-2B KnowSelf matches GPT-4o ExpeL

Interpretation: KnowSelf shifts the cost model from “always retrieve” to “retrieve only in knowledgeable-thinking states.”

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1

Indiscriminate knowledge injection is wasteful and can be suboptimal.

2

Agents can learn situational self-awareness via lightweight heuristic labeling and special tokens.

3

Llama-8B KnowSelf reaches 84.33%, outperforming GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).

4

Only 15–26% knowledge usage is needed versus 100% for full-knowledge competitors.

5

The paradigm scales down: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o ExpeL.

Future directions

- Richer token taxonomies beyond three modes
- Adaptive retrieval budgets and latency-aware planning
- Transfer to open-ended web, tool, and embodied tasks
- Robustness under noisy or incomplete knowledge bases

Act · Reflect · Retrieve